SPMM Smart Revise

Adult Psychiatry I

Paper B **Syllabic content 7.1**

1. Major Depressive Disorder

Epidemiology: Various studies including NCS and its replication NCS-R, NSEARC and ECA have **implicated that the lifetime prevalence of depression is changing**. The 1-year prevalence of depression in the general population was 5.3% while lifetime prevalence was 13%. NSEARC showed that the point prevalence estimate of depression nearly doubled from 3.3 to 7% (1992 to 2002). The **highest risk was for age group greater than 30**, as opposed to previous finding of 19 to 29 age group.

According to NESARC (National epidemiological survey of alcoholism and related conditions. Hasin et al. Arch. Gen. Psychiatry 2005; 62, 1097) **mean age of onset of depression is 30 years**. The first depressive episode occurs before age 40 in about 50% of patients. The mean **number of episodes in patients with lifetime major depressive disorder is 5**. The **longest duration was 24 weeks** per episode.

According to EseMED (European data), only 21% of depressed patients received antidepressants in a year. One-third of identified patients have consulted their general practitioner in preceding 12 months while 21% cases have seen a psychiatrist. Nearly one-third of those who sought help, have never seen any mental health professional. Nearly 21% remained untreated in spite of seeking help. The mean age of treatment onset 33.5 years – lag 3 years.

Most **common comorbidity was alcohol** use (>40%) and anxiety (>40%), 30% had a personality disorder. Notably, 29% of those who have alcohol dependence satisfy criteria for the 1-year prevalence of clinical depression. Cluster C personality disorders except anankastic PD show strong associations with lifetime MDD. **Suicide is attempted by 9%**.

According to DEPRES (Depression Research in European Society) study, a significant proportion of sufferers from depression **(43%) failed to seek treatment** for their depressive symptoms. **Sufferers from major depression made almost three times as many visits to their GP** or family doctor as non-sufferers, placing huge demands on the primary care. Only 31%of those who sought help were given a drug therapy and among these, only 25% were given antidepressant drugs

Diagnostic stability: In approximately 56% of patients, the initial diagnosis of depressive disorder eventually changes during follow-up mainly to the schizophrenic spectrum (16%), but also to personality disorders (9%), neurotic, stress-related and somatoform disorders (8%) and to bipolar disorder (8%) (Kessing, 2005).

In community studies, one in ten patients who begin with a depressive episode go on to develop an episode of mania within 10 years. If the illness began at a younger age, the switch was earlier. This rate increases to nearly 50% if severely depressed hospitalised patients are considered. Among hospitalised depressed patients followed up for nearly a decade 1% a year converted to bipolar I and 0.5% a year converted to bipolar II.

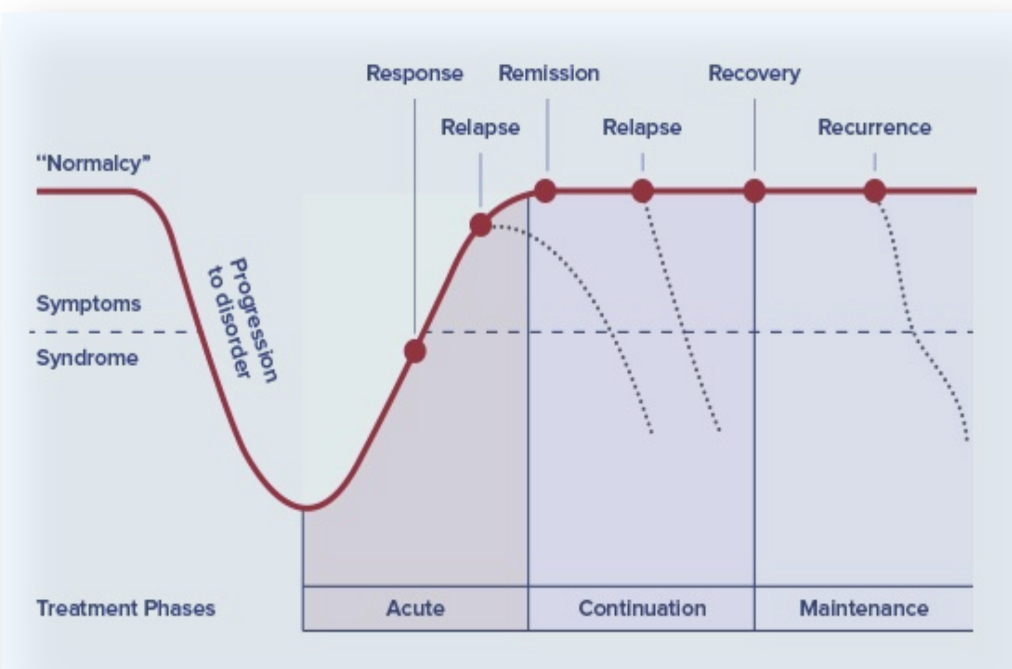
It is known that the majority of bipolar patients, particularly women, begin with depressive episodes. Factors associated with a change of polarity from unipolar to bipolar were younger age, family history of bipolarity, antidepressant-induced hypomania, hypersomnia and retarded phenomenology, psychotic depression, and postpartum episode. The mean age at which the switch occurs is 32 years. The average number of previous episodes in those who switch varies between two and four.

Mortality/ physical morbidity: Longstanding or recurrent depression is a significant predictor of mortality (RR=1.5, 1.06-2.20). This risk may come down in those who were once depressed but recovered now. People with major depression have up to a five-fold increased risk of MI. Taking all diagnostic categories together, suicide among mood disorder patients was nearly 14 times more frequent than in the general population.

Duration of episodes: An **untreated depressive episode lasts 6 to 13 months**; most **treated episodes last about 3 months**. As the course of the disorder progresses, patients tend to have more frequent episodes that last longer.

Recurrence rates: In a population-based follow-up study of first episode depression, the median episode length was noted to be 12 weeks. About 15% of those with first episodes did not have a year free of episodes, even after 23 years. About 50% of first episode participants recovered and had no future episodes. Approximately 50% of patients with an episode of depression will experience a recurrence within 5 years. For those who have had 2 major depressive episodes, the probability of recurrence goes up to 70%, and after three episodes it increases to 80% to 95%. Paykel et al showed that the risk of relapse or recurrence is significantly higher in patients with residual symptoms (75% in 15 months) after treatment compared with patients who were treated to full remission (25% in 15 months).
Over a lifetime, bipolar patients experience twice as many episodes as unipolar depressives

The 5 Rs of depression & The Kupfer Curve: Imagine someone in depression; if they achieve 50% symptom-free they are **responding** to treatment; 26-49% decrease in symptom severity compared with baseline is termed as partial response; once they achieve a state where no scales can detect meaningful measure of depression (say HAMD<7) and continue to do so even after the natural period of a treated depressive episode (usually >3m) is over, they are said to be in **remission**. If they continue to be remitted after the natural period of an untreated depressive episode (usually



>6m after the acute period), they are said to be **recovered**. Any repeat of depressive episodes before this period is a **relapse** and after this is a **recurrence** i.e. a new episode. **(**From www.rethinkdepression.com)

Good prognostic indicators	**Relapse indicators**
Mild episodes	Persistent dysthymia
Absence of psychotic symptoms	Comorbid conditions, including both psychiatric and general medical conditions.
A short hospital stay	Female sex
History of solid friendships during adolescence.	Longer episodes of illness before seeking treatment,
Stable family functioning.	Greater number of prior episodes (3 or >),
Sound social functioning for the 5 years preceding the illness.	Never marrying
The absence of the comorbid psychiatric disorder.	Remission status at 3 months – partial remission predicts recurrence
No more than one previous hospitalization for major depressive disorder.	Previous episode in past year
	Severity of episodes (for example, suicidality and psychotic features)
	Long previous episodes

Evidence-Based Treatment:

NICE guidelines for depression - summary:

NICE guidelines for depression - summary

- Most points are for primary care management.
- Screening high-risk primary care attendees (past history, physically ill, demented, etc.) is necessary
- If comorbid depression and anxiety present, treat the depression first.
- Depression must be classified as per severity for proper management
- Watchful waiting with no active intervention for mild depression is an advocated strategy if agreeable; the review must be in 2 weeks.
- Antidepressants have a poor risk-benefit ratio for mild depression – so not advocated. CBT-based guided self-help can be advised.
- For mild/moderate depression CBT, counselling or problem-solving therapy can be advised.
- SSRIs to be used as the first line of antidepressants.
- When patients present initially with severe depression, a combination of antidepressants and CBT should be considered as the combination is more cost-effective than either treatment on its own.
- For patients with a moderate or severe depressive episode, continue antidepressants for at least 6 months after remission
- Patients with >2 episodes in recent past, or with residual impairment should continue antidepressants at least for 2 years
- SSRIs to be used first in atypical depression before referral to a specialist. In women with atypical features, the specialist may consider phenelzine if no response to SSRIs.
- Patients who have had multiple episodes but good response to lithium augmentation of antidepressant should remain on the combination for at least 6 months
- Electroconvulsive therapy (ECT) should only be used to achieve rapid and short-term improvement of severe symptoms after an adequate trial of other treatments has proven ineffective, and/or when the condition is considered to be potentially life-threatening, in a severe depressive illness.
- ECT maintenance is not recommended.

<u>Principles of prescribing in depression</u>

- For a single episode, continue treatment for at least 6-9 months after resolution of symptoms (multiple episodes may require longer)
- Withdraw antidepressants gradually; always inform patients of the risk and nature of discontinuation symptoms
- <u>SSRIs - Summary of Suggestions from the Medicines and Healthcare products Regulatory Agency:</u> Use the lowest possible dose. Monitor closely in early stages for restlessness, agitation and suicidality. This is particularly important in young people (<30 years). Doses should be tapered gradually on stopping.

<u>Do antidepressants work?</u>

Reanalysis of individual patient data available at FDA has shown the following;

- In adults NNT for an antidepressant response seems to be 4-5; for remission it may be 6-7 for short-term trials (around 12-14 weeks). In elderly it is found that the NNT for tricyclics is 4; 8 for SSRIs and 3 for MAOIs. In children and adolescents NNT for SSRIs was of uncertain clinical and statistical significance; paroxetine had NNT benefit 12 (5 to ∞) and NNT harm 20. Sertraline had NNT benefit 10 (95% CI 5 to ∞). Fluoxetine had somewhat clinically worthwhile NNT benefit of 5 (4–13).
- **Kirsch's meta-analysis** of data from 47 trials (conducted between 1987 and 1999, and submitted to FDA for successfully approved antidepressants) included 4-8 weeks RCTs of nefazodone, venlafaxine, fluoxetine and paroxetine. The weighted mean improvement was 9.60 points on the HRSD (Hamilton) in the drug groups and 7.80 in the placebo groups, yielding a mean drug–placebo difference of 1.80 on HRSD improvement scores. This difference attained statistical significance but did not meet the three-point HRSD criterion difference between drug–placebo required by NICE for clinical significance. But the magnitude of the difference (effect size) was a function of baseline severity of depression. The effect of placebo declined as severity increased. This significant difference between drug and placebo (i.e. exceeding NICE's 0.50 standardized mean difference criterion) was observed in comparisons exceeding HRSD score 28 in baseline severity.
- **<u>Stepped care approach:</u>** NICE advocates a stepped care model advocated by WHO in managing chronic illnesses.

Primary care and General Hospitals - Recognition
- Assessment & Screening

Treatment of mild depression in primary care
- Mild depression - watchful wait, self-help, computerised CBT, exercise, brief psychological therapies.

Treatment of moderate-severe depression
- Medication use, psychosocial support

Specialist treatment
- Medication, omplex psychotherapies, combined treatments

Inpatient treatment
- Risk management - ECT/combined treatments

Phases of depression treatment: from Hirschfeld 2001 **(after one episode)**

Acute phase	Stabilisation of acute symptoms – up to 3 months
Continuation phase	Provided relapse free, this lasts 6-12 months. Covers the natural (if untreated) course of depression.
Maintenance phase	Aims to prevent recurrences – depends on risk factors and probability or recurrence.

To continue or not? Geddes and colleagues performed a pooled analysis of data from 31 randomized trials that included 4,410 patients; they found that continued treatment with all classes of antidepressants reduced the risk of relapse by 70% compared with treatment discontinuation after acute episode, which was a highly significant difference. The average rate of relapse on placebo was 41% compared with 18% on active treatment. The treatment effect seemed to persist for up to 36 months, although most trials were of 12 months' duration.

STAR*D

1. Sequenced treatment alternative for depression was a p*ragmatic RCT* study – nearly 2/3rd had comorbid physical disorder, nearly 2/3rd has co-morbid psychiatric diagnosis and about 40% had onset of depression less than age 18 – very similar to real world findings.
2. Enrolled *4041 patients* with a broad range of symptoms and severity at 25 participating sites in the USA.
3. The study involved 4 possible 'steps' for treatment, and any patient who failed to meet remission criteria at each step was then asked to move to the next level. In each level, more than one randomisation options were available
4. **Level 1:** Citalopram was given (n = 3671; dropout rate was approximately 8%) - Patients were encouraged to continue the treatment for up to 12 weeks
5. **Level 2**: After 12 weeks, if patients failed to reach remission in level 1, they were randomized to the next level, depending on their preference to 'switch' (bupropion SR, N = 239; sertraline, N = 238; or venlafaxine XR, N = 250), switch to cognitive therapy (N = 62), 'augment' citalopram (bupropion SR, N = 279; or buspirone, N = 286), or 'combine' citalopram with cognitive therapy (N = 85).
6. **Level 3:** Participants who did not achieve remission after 12 weeks in level 2 were randomized to switch to mirtazapine (N = 110); switch to nortriptyline (N = 116); or augment level 2 treatment with lithium (N = 63) or thyroid medication (N = 70).
7. **Level 4**: Patients who did not achieve remission after 12 weeks in level 3 were required to switch to the monoamine oxidase inhibitor, tranylcypromine (N = 55); or switch to a combination of venlafaxine XR and mirtazapine (N = 50).
8. To ensure that every participant had the best chance of recovery with each treatment strategy, a systematic approach called **measurement-based care** was used. Accordingly, routine measurement of symptoms and side effects at each treatment visit with the use of a treatment manual-guided when and how to modify doses tailored to each individual.
9. Patients tended to choose an option based on their experiences with the initial antidepressant. If they experienced a partial response, they chose augmentation; if they were not responding, they preferred to switch, and so forth.

10. Remission rates dropped while relapse rates increased with moving higher to each level (see table below)

STAR*D	Level 1 (SSRI mono)	Level 2 (switch or augment or combine CBT)	Level 3 (switch MZP/NTP or Li or T3 augment)	Level 4 (switch MAOI or VFX/MZP combo)
Remission (Quick Inventory of Depressive Symptomatology–Self-Report)	37%	31%	14%	13%
Relapse rates (more in those who did not have complete remission)	34%	47%	43%	50%
Intolerance to treatment	16%	20%	26%	34%
Drop-out rates	After level 1 21% withdrew	After level 2, 30% withdrew	After level 3, 42% withdrew	

11. The *cumulative remission rate* after all 4 steps was 67%; the cumulative non-response rate was 33%. About half of participants in the STAR*D study became symptom-free after two treatment levels.
12. The observed differences among the *3 switch options* -- sertraline, bupropion SR, and venlafaxine XR -- were small and below the threshold of statistical significance.
13. *Switch to a class* outside SSRIs was not different from the within-class switch option.
14. At level 3 again no statistical difference was seen within 2 different switches or 2 different augmentations (switch vs. augment not compared as patients chose one or other – were not randomized). T3 was better tolerated than lithium.
15. At level 4 again no differences were noted between MAOI and VFX/MZp combination though the degree of symptom relief (not proportion responding) was better with VFX/MZp combo.

<u>Who will drop out?</u>

In STAR*D immediate attrition after one visit was associated with younger age, less education, and higher perceived mental health functioning. Attrition later in treatment (but within 12 weeks) was associated with younger age, less education, and African American race. Experience with more than one episode of depression was associated with less attrition.

<u>Does dose escalation help non-responders?</u>

At least for SSRIs the evidence is equivocal. In STAR*D dose escalation was not used as an individual option or step in managing depression. A systematic review identified 8 true dose-escalation studies and 3 meta-analyses. The available data provided no unequivocal base for dose escalation. Especially dose escalation before 4 weeks of treatment at a standard dose appeared to be ineffective.

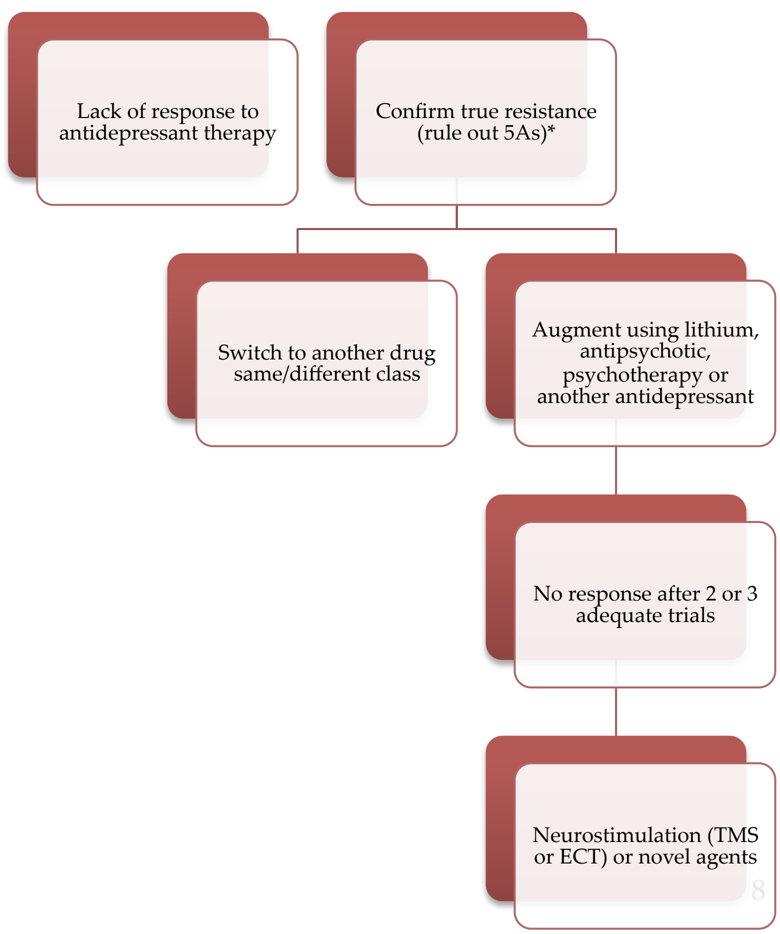
Also, when the patients and clinicians make the choice of antidepressant randomly, significantly lower symptom reduction occurs, when compared to the algorithm-based treatment of depression (Trivedi et al., 2006).

Are there any gender differences?

Men report more suicidal ideation. But women report more suicidal attempts. Men are 2-4 times more likely to be successful in their suicidal attempts. Women have more symptoms of anxiety and atypical depression, earlier age of onset of depression and a trend towards longer episodes. Men have more psychomotor agitation and substance use than women. According to BAP guidelines (Cleare et al., 2015), "there is no consistent evidence for a clinically important effect of gender on response to different antidepressants, although younger women may tolerate TCAs less well than men.

Do antidepressants increase suicide risk?

- MHRA has issued a black box warning that Under 18s may have an increased risk of suicide secondary to SSRI prescription. This increased risk may be seen in adults too. In May 2007, the Food and Drug Administration (FDA) ordered that all antidepressant drugs carry an expanded black-box warning incorporating information about an increased risk of suicidal symptoms in young adults aged 18–24 years.
- The risk of suicidal behaviours associated with antidepressants was found to be elevated in subjects under age 25, neutral in subjects aged 25–64 (reduced if suicidal behaviour and ideation are considered together), and reduced in subjects aged 65 and older (so protective in elderly).
- Khan, analysing FDA data concluded that no increase in suicides is seen in adults, but Healy reappraised this and showed that ***1.62 times more suicides*** mean a significant difference.
- BAP's most recent update of antidepressant use (2015) states that "Antidepressant (including SSRI) treatment is not associated with an increased risk *of completed suicide* and ecological studies find it is associated with decreased suicide rates. Antidepressant (including SSRI) treatment does not appear associated with a clinically significant increased risk of suicidal behaviour in adults, although the individual sensitivity cannot be ruled out. SSRIs are associated with a small (<1%) increase in non-fatal suicidal ideation/behaviour in adolescents/younger adults with a benefit–risk ratio of >10".



- From epidemiological and cross-sectional data, it is likely that suicidal behaviour leads to treatment rather than treatment leading to suicidal behaviour.
- **There are several difficulties in studying the association between suicide and antidepressant usage.** Some results such as sertraline increases self-harm but not suicide risk do not make ***biologically plausible sense***. Suicide is a ***complex behaviour*** and studying it in 'all or none' manner is not useful. The role of ***anxiety, akathisia*** induced by SSRIs in this is unclear. Nevertheless, the reported association may not be a ***general psychotropic prescription effect***; prescription related suicides are not seen with antipsychotic usage (Khan, 2001).

Antidepressant toxicity in overdose

In an observational study by Hawton et al. 2010, TCAs showed greater toxicity than venlafaxine and mirtazapine, both of which had greater toxicity than SSRIs. Within the TCAs, dosulepin and doxepin were more toxic in comparison with amitriptyline. Within the SSRIs, citalopram had a higher case fatality than other SSRIs.

Managing treatment resistant depression (TRD):

The 5As that can result in apparent resistance to antidepressant treatment ('pseudoTRD') are **Alcoholism,** lack of **Adequate dosage,** lack of **Adherence, Axis 2 disorders** (personality disorders), **Alternate diagnoses** (e.g. PTSD). See the flowchart above for further details.

Is combining two antidepressants an effective strategy?

The evidence base for combination antidepressants is very sparse (Palaniyappan et al., 2008). There are no clear recommendations that could be made from available evidence, but the following points are for the purpose of MRCPsych MCQs:

1. Serotonin syndrome is a risk with many combinations especially SSRI-MAOIs
2. Venlafaxine mirtazapine combination is called Californian rocket fuel. This has some support for use in TRD
3. The neuropharmacological rationale must be checked before combinations – for e.g. combining two SSRIs may not give any pharmacodynamic advantage, combining two different mechanisms of action must be aimed for.
4. SSRI TCA combinations may work both by pharmacodynamic and kinetic mechanism – SSRIs inhibiting TCA metabolism.
5. Some combinations may be used for specific symptom relief e.g. trazodone for insomnia in combination with other antidepressants.
6. No single combination is found superior to others in head to head trials.

Novel antidepressant options: (Palaniyappan & McAllister-Williams, 2007)

Agomelatine is a new antidepressant that is a $5HT_{2C}$ antagonist (and a melatonergic agonist). GABA interneurons tonically inhibit noradrenergic circuits (from locus coeruleus) and dopaminergic circuits (from ventral tegmentum) projecting to the prefrontal cortex. $5HT_{2C}$ receptor stimulation drives these GABA interneurons. Thus, norepinephrine and dopamine circuits are inhibited by the normal tonic release of serotonin (Stahl, 2007). Agomelatine, through $5HT_{2C}$ inhibition, therefore, may in part act as a norepinephrine and dopamine disinhibitor. Clinical trials have confirmed significant antidepressant efficacy for this agent (Olie´ and Kasper, 2007) though the relative contribution of melatonergic properties of this drug still remains unknown.

Side effects: Combining benzodiazepines with antidepressants early in treatment speeds response and reduces drop-outs and may be useful for managing early agitation/anxiety and insomnia, but needs to be balanced against the risk of long-term use.

Beneficial strategies for sexual side effects are: switching to an anti- depressant with a lower tendency to cause sexual side effects (II); adding sildenafil (I) or tadafinil (II) for erectile dysfunction, or bupropion 150 mg twice daily for sexual dysfunction (II), in men; and adding bupropion (I) or sildenafil (II) in women. Modafinil may improve sleepiness in partial respond- ers to SSRIs with fatigue and sleepiness but its effect on fatigue is unclear (II).

<u>What about St. John's wort?</u>

An initial meta-analysis of St Johns's Wort (Hypericum perforatum) published in 2001 showed an adjusted RR of 1.94 (1.5 to 2.5) in favour of SJW. But when 3 later studies were added, and the data was re-analysed, this effect size dropped to 1.30 (1.0 to 1.60). In fact, SJW may not be as effective as once claimed. Evidence from at least one systematic review of multiple well-designed RCTs are found for the use St John's wort, tryptophan/5-hydroxytryptophan, *S*-adenosyl methionine, inositol and folate in depressive disorders. None of these findings was conclusively positive, and folate had a significant effect only when combined with an antidepressant. St John's wort can increase the effects of conventional SSRIs.

Ketamine:

- The anaesthetic and hallucinogenic drug
- Demonstrated to be an extremely rapidly acting antidepressant acting via blockade of glutamatergic NMDA receptors and relative upregulation of AMPA receptors. May also act on mammalian target of rapamycin (mTOR) and brain-derived neurotrophic factor (BDNF to affect intracellular signalling.
- The clinical evidence from relatively small trials to date reveals rapid improvements in mood and suicidal thinking in most participants (~70%), although the effects appear temporary when a single infusion is administered, with no pharmacological augmentation effective to date.
- Ketamine (0.5 mg/kg single dose) demonstrated a notable improvement over placebo saline at 24 hours (effect size = 1.46; later dropping to 0.68 by 1 week; 71% on ketamine showed response) (Zarate et al., 2006). Two studies investigating the effect of ketamine in bipolar depression yielded similar positive results [DiazGranados et al. 2010b; Zarate et al. 2012]. Studies have evaluated augmenting ECT with a sub anaesthetic dose of ketamine, using ketamine as the anaesthetic agent.

2. Bipolar Disorder

Epidemiology

- Consistently in large epidemiological studies, 1.5% point prevalence is quoted for bipolar disorders. But note that hypomania, being positively appraised by patients themselves, is often missed in structured, non-clinician interviews, so underrepresented in surveys.
- In NCS-replication (part of World Mental Health survey initiative), **lifetime prevalence of bipolar 1 was 1.0%, bipolar 2 was 1.1%** while subsyndromal bipolar spectrum was 2.4%.
- NCS-R showed that the mean age of onset for BP-I was 18.2 while BP-II started around age 20. Subsyndromal BPD started around 22 years of age.
- Clinical severity and role impairment were more common in BP-II than BP-I (Merikangas et al.; Arch Gen Psychiatry. 2007; 64:543-552). Angst et al. (2003) observed in a 20-year long prospective study that patients with depression and clinically undiagnosed subsyndromal hypomania have similar risk factors, course and outcome compared to bipolar disorder type 2.
- **The suicide rate in bipolar disorder is ~15-18 times higher** than the general population.
- According to NICE, around two-thirds of people with bipolar disorder also experience another mental disorder, usually anxiety disorders, substance misuse disorders, or impulse control disorders (Kendall et al., 2014))

The spectrum

Akiskal, Angst and other authors have long supported the view of a spectrum of mood disorders. This spectrum can be expressed in two dimensions – a **proportionality** dimension ranging from predominant depression to predominant mania and a **severity** dimension ranging from normal day-to-day emotional states to psychotic mood states.(adapted from Angst 2007)

Severity	Predominant depression → ________→		Predominant mania		
Psychotic	Psychotic unipolar depression	Bipolar 2; psychotic depression	Bipolar 1 both depression and mania - psychotic	Mania (psychotic) and mild depression	Unipolar mania - psychotic
Non psychotic major mood disorder	Non psychotic unipolar depression	Bipolar 2; non-psychotic depression	Bipolar 1 both depression and mania – non-psychotic	Mania (non-psychotic) and mild depression	Unipolar mania no psychosis
Subthreshold/minor	Dysthymia, recurrent brief depression		cyclothymia		hypomania
Personality disorders	Characterological depression – depressive poersonality		Borderline personality (affective instability)		
temperament	Depressive temperament		Cyclothymic temperament		hyperthymic temperament
Symptoms (normal)	sadness				happiness
No symptoms (supernormal)					

Akiskal and Pinto (1999) described an extension to bipolar 1 & 2 classification.

Evolving Spectrum of Bipolar Disorders
Bipolar I: Manic-depressive illness
Bipolar I 1/2: Depression with protracted hypomania
Bipolar II: Depression with spontaneous hypomanic episodes
Bipolar II 1/2: Depression superimposed on cyclothymic temperament
Bipolar III: Recurrent depression, plus hypomania occurring solely in association with antidepressant or other somatotherapy
Bipolar III 1/2: Mood swings that persist beyond stimulant and/or alcohol abuse
Bipolar IV: Depression superimposed on a hyperthymic temperament

Note that DSM-V now includes an emphasis on changes in activity and energy as well as mood, when making a diagnosis of manic and hypomanic episodes.

Is there a prodrome for bipolar disorder?

A predominantly insidious prodrome (>1 year) is seen in more than half of the patients; acute presentations with less than 1 month of prodrome were very rare; subacute was more common than acute but less common than insidious prodromes. The prodrome duration did not differ between those with psychotic and nonpsychotic mania. Attenuated positive symptoms and increased energy/goal-directed activity were significantly more common in the prodrome of patients with later psychotic mania. For mania and schizophrenia, the prodromal characteristics overlapped considerably. However, subsyndromal unusual ideas were significantly more likely part of the schizophrenia prodrome; obsessions/compulsions, suicidality, difficulty thinking/communicating clearly, depressed mood, decreased concentration/memory, tiredness/lack of energy, mood lability, and physical agitation were more likely part of the mania prodrome.

Ambelas confirmed that there was a strong correlation between stressful life events and first manic admissions. This lessens as the illness progresses. This is particularly true for younger bipolar patients and is significantly linked to mania and not depression. Sleep reduction may be a final common pathway to mania.

Prognosis of bipolar disorder:

Diagnostic stability: Approximately 40% of patients with bipolar disorder get misdiagnosed previously as having unipolar depressive disorder initially. But once a diagnosis is made the stability of bipolar disorder has generally been reported to be high, ranging from 70% to 91%. In bipolar disorder, bipolar I is more stable than bipolar II just because bipolar II may switch to bipolar I, but not vice versa.

Course: Median time to recover from mania (treated) is around 4-5 weeks. The course specifiers in DSM include chronicity (with or without full interepisode recovery), seasonality, and rapid cycling. Around 56% bipolar patients display a specific pattern of predominant polarity; 60% of those may be classified as predominantly depressed (with at least two-thirds of past episodes fulfilling criteria for major depression), whereas 40% may be classified as predominantly manic or hypomanic.

Depressive polarity	Manic polarity
Around 1/3rd of bipolar patients	Around 1/4th of bipolar patients
More in bipolar 2	More bipolar 1
Onset is with depressive episode often; somewhat later age	Onset is often with mania; earlier age
More seasonality	Less seasonal;
More suicide attempts	More substance use
Better response to Lamotrigine on longer term	Better response to antipsychotics in long term
More exposure to antidepressants	

(From Vieta & Phillips, 2007)

Mortality: Suicide rates in people with bipolar disorder is estimated to be around 10% - 19%. This is 15 times that of the general population At least a quarter will attempt suicide. This is higher than that seen in unipolar depression. The depressed phase of bipolar disorder is linked to about 80% of suicide attempts and completed suicides. Mortality rates from natural causes (mainly vascular disease) are also increased in people with bipolar disorder (Mitchell et al., 2004)

Recurrence & recovery: .According to NICE, the risk of recurrence in the year after a mood episode in patients with bipolar disorders is especially high (50% in one year and >70% at four years) compared with other psychiatric disorders (Kendall et al., 2014)

Bipolar 1 vs. 2: Bipolar II disorder is often thought of as a milder expression of bipolar disorder. However, it is only milder in its severity of manic symptoms as hypomania is less severe than mania. Patients with bipolar type II disorder suffer more depressed days than patients with bipolar type I disorder over time. According to prospective studies, bipolar II patients experienced mood symptoms 54% of the time compared with 47% in bipolar I patients. For both bipolar I and bipolar II patients, those mood symptoms were dominated by depressive versus manic symptoms, although bipolar II patients experienced more depressive symptoms (93%) than bipolar I patients (67%). This finding may explain the increased suicidality rates and poor functional recovery observed among bipolar 2 patients (Keck, 2007).

Relapse indicators:

- Perlis et al. (STEP-BD) found that the biggest predictor of relapse was ***residual symptoms***, and the majority of relapses were into depression.
- ***Index phase*** may have a bearing on relapse and overall outcome. Patients with a depressive index phase spent the highest proportion of time being ill (~65% of the time), especially with depression in both bipolar 1 and 2 types. Patients with a manic index phase spent the smallest proportion of time ill (30% time nearly), especially in depressive states (n= 191, 18m follow up. Mantere et al., 2008).
- ***Sleep disruption*** is often the 'final common pathway' triggering mania and depression: ***stressors*** that lead to reduced sleep may contribute to relapse. Sleep disturbance may also be a sign of relapse having occurred and a hypomanic state having set in. Social and circadian rhythm disruptions are strongly linked to bipolar relapses.

- ***Comorbidities*** often contribute to relapses. A current substance use disorder increases the risk of manic relapse while a current anxiety diagnosis was associated with increased risk of a depressive relapse but not manic relapse (Perlis, 2005). Alcoholism is the commonest significant clinical co-morbidity in Europe.

Switch:

Antidepressant-induced mania is called as 'switch'. This is a short-term phenomenon and takes place soon after starting antidepressants (Ghaemi empirically defines it as happening within 2 months of antidepressant treatment). '***Mood destabilization***' is a long-term phenomenon wherein antidepressants result in increased frequency of mood episodes over time than would have occurred in the natural course. Antidepressants may cause long-term mood destabilization without a short-term manic switch, and vice versa. In spite of having low rates of acute manic switch, new generation of antidepressants can produce long-term mood destabilization – as suggested by the results of STEP-BD trial (Ghaemi, 2008)
Antidepressant-induced mania/hypomania has been reported with all major antidepressant classes in a subgroup of about 20-40% of bipolar patients. Lithium may confer better protection against this outcome when compared with other standard mood stabilizers, although switch rates have been reported with comparable frequencies on or off mood stabilizers. Mood switches were less frequent in patients receiving lithium (15%) than in patients not treated with lithium (44%) in a small naturalistic study.
Risk factors for antidepressant-induced switch includes

(i) previous antidepressant-induced manias,
(ii) a bipolar family history, and
(iii) exposure to multiple antidepressant trials and
(iv) initial illness beginning in adolescence or young adulthood.

The switch can also occur from mania to depression when treated with antimanic agents. Typical antipsychotics are more likely to result in patients switching from mania into depression. Evidence in this area is very limited, but BAP guidelines quote data from the lamotrigine/ lithium/ placebo trials suggesting that the risk of relapse of the index episode was higher than the risk of switching during treatment.

Differentiating bipolar and unipolar depression:

Very few significant differences have been noted, but these are not consistent across studies. It is thought that bipolar depression has a shorter natural course though this is not clearly proven.

Unipolar depression	Bipolar depression
More anxiety	Lesser anxiety
More somatic complaints	Fewer physical complaints
Less withdrawal	More withdrawal
Lesser degree of retardation	More retardation
Insomnia	Hypersomnia
Less atypicality	More atypical symptoms

Psychomotor differences have largely been identified in inpatient studies, and may not be as useful among less severely ill outpatients. Higher family history of bipolar disorder suggests bipolar depression; the presence of psychotic depression in early adulthood is also suggestive of bipolar disorder.

Rapid cycling:

The phenomenon of rapid cycling involves experiencing **≥4 episodes/year**. This includes **both depression and mania**. Rapid cycling can appear and disappear at any time during the clinical course of bipolar disorder. Patients who have ≥4 episodes/month are considered ultra-rapid cyclers and those who switch within a day on ≥4 days/week are termed ultra-ultra rapid or ultradian cyclers. 20% of bipolar patients had rapid cycling in STEP-BD cohort. Women were more likely to be rapid cyclers than men (nearly 80% of all rapid cyclers are women). Rapid cyclers had an earlier onset of illness than non-rapid cyclers. Rapid cyclers had more severe depression and mania and had lower global functioning (Schneck et al 2004). It is controversial whether bipolar 2 patients have a higher proportion of rapid cycling. STEP BD reported both bipolar 1 and 2 having an equal incidence of rapid cycling, but Kupka (2003) in a meta-analysis reported that bipolar 2 patients have more rapid cycling spells.
Current hypothyroidism and poor response to lithium (especially the depressive component) are also associated with rapid cycling.

Secondary mania:

Secondary mania due to organic brain damage (especially right hemisphere) is more common in the elderly. L-Dopa and corticosteroids are the most common prescribed medications associated with secondary mania. In stimulant or other street drugs induced mania, if the mood state significantly outlasts the drugged state then a diagnosis of bipolar disorder can be made. Hypothyroidism creates a picture akin to depression; in hyperthyroid state one may present as hypomanic or agitated depressed.

Mania vs. hypomania:

Note that the difference between mania and hypomania lies in the degree of functional impairment. The need for hospitalisation can be considered as a proxy of functional deterioration. The duration criteria of 4 days for hypomania and 7 days for mania in DSM is quite arbitrary; follow-up studies have shown that most hypomanic episodes in bipolar 2 lasts for less than 4 days.

Evidence-Based Treatment (as per NICE CG185, updated 2014 and BAP recommendations)

Treatment of acute de novo mania:

- Antipsychotics (haloperidol, olanzapine, quetiapine, or risperidone) must be used first line in the previously untreated because of their rapid anti-manic effect. . Do not offer lamotrigine to treat mania (New recommendation).
- If the patient is already on an antidepressant monotherapy: Consider stopping the antidepressant first; then offer an antipsychotic regardless of whether the antidepressant is stopped. (New recommendation.)
- Adjunctive benzodiazepine, such as clonazepam or lorazepam can be used for agitation and insomnia. RCTs in people with bipolar type I disorder experiencing a manic episode suggest that clonazepam may be as effective as lithium in improving manic symptoms at 1–4 weeks. But guidelines do not recommend it as monotherapy.

Acute manic relapse in a known bipolar patient:

- Increasing the dose of mood stabiliser must be the first option.

- Check serum lithium levels and consider establishing a higher serum level if compliance is good. If the person is already on lithium, optimise plasma levels first; then consider adding haloperidol, olanzapine, quetiapine, or risperidone
- Antipsychotic augmentation can also be done for patients on valproate.
- ECT may be considered for severely ill manic patients, treatment-resistant mania, for those who prefer ECT and patients with severe mania during pregnancy.
- For psychosis during a manic or mixed episode that is not congruent with severe affective symptoms, antipsychotics must be used.

Carbamazepine is associated with a risk of hyponatraemia – often higher than other psychotropics. This is related to SIADH. In some cases, carbamazepine has been used to treat diabetes insipidus (not when it is lithium related). Oxcarbazepine, a related drug, has 3-5% risk which is even higher than carbamazepine.

Treating bipolar depression:

- NICE suggests offering a psychological intervention specific for all bipolar depression or high-intensity psychological intervention for depressive disorder as the first line.
- NICE also states that "If a person develops **moderate or severe** bipolar depression and is not taking a drug to treat the disorder, offer fluoxetine combined with olanzapine, or quetiapine on its own, depending on the person's preference and previous response to treatment". The second step is offering Lamotrigine. Any of the three can be used (including olanzapine on its own, but not fluoxetine on its own) if the patient prefers so.

Maintenance treatment:

- **BAP guidelines recommend** long-term treatment (relapse prevention/maintenance) following a single severe manic episode (i.e. diagnosis of bipolar I disorder). Long-term treatment even if does not control all relapses, can make them less severe.
- **NICE recommends** using long-term maintenance treatment
  - after a manic episode involving significant risk and adverse consequences
  - bipolar I disorder with two or more acute episodes
  - Bipolar II disorder with significant functional impairment, or risk.
- Lithium monotherapy is the first line. It is probably effective against both manic and depressive relapse, although it is more effective in preventing mania. Lithium is also associated with a reduced risk of suicide in bipolar disorder.
- Lithium maintenance responders' profile: Euphoric mania, no rapid cycling, full interepisode remission, no comorbidity, no psychotic features, fewer lifetime episodes, mania-depression-euthymia course.
- Other options include:
- Valproate - probably prevents manic and depressive relapse
- Olanzapine- prevents manic more than depressive relapse
- Quetiapine – especially if it was effective during an acute episode
- Carbamazepine - less effective than lithium but may be used as monotherapy
- Lamotrigine - prevents depressive more than manic relapse.

Managing mixed episodes:

- Must be treated as manic episodes. Avoid antidepressants if possible. Higher risk of suicide may be present. No single agent with clear superiority in mixed episodes. The best evidence for efficacy in treating mixed episodes is for valproate, which has been shown to be more effective than lithium or placebo (Mitchell et al., 2004).
- Atypical antipsychotics have also been shown to be effective. Monotherapy may be insufficient, and patients may require combination treatment. Subgroup analysis of olanzapine vs. placebo in prophylaxis for bipolar revealed olanzapine treatment significantly lengthened the time to relapse in patients with a mixed index episode. For this mixed subgroup, estimated median time to bipolar relapse was 46 days for olanzapine and 15 days for placebo. But no specific recommendations can be made using this secondary analysis.

Managing rapid cycling:

- Hypothyroidism or substance misuse may contribute to rapid cycling. These must be managed.
- Antidepressants may contribute and so should be discontinued.
- Suboptimal medication regimes, the effects of lithium withdrawal, and erratic compliance must also be considered when treating rapid cycling (NICE)
- For initial treatment of rapid cycling lithium, valproate or lamotrigine could be considered.

NICE guidelines for bipolar disorder (2014 update) - summary:

NICE guidelines for bipolar disorder – summary
➢ Encourages routine enquiry about hypomania in those who see GPs with depression. ➢ Avoiding excessive stimulation – delaying important decisions– calming activities – a structured routine with a lower activity level must be advised for all patients during an acute manic episode ➢ Rapid cycling episodes must be managed in secondary care ➢ Avoid antidepressants for patients who have rapid-cycling bipolar disorder, a recent hypomanic episode and recent functionally impairing rapid mood fluctuations. ➢ For mixed episodes consider treating patients as if they had an acute manic episode, and avoid prescribing an antidepressant. ➢ Maintenance should be continued for at least 2 years after an episode of bipolar disorder or up to 5 years if the person has high-risk factors for relapse. ➢ In paediatric bipolar disorder, use as first line an atypical antipsychotic to treat mania.

Interestingly though BAP and NICE endorse adjunctive antidepressant use in bipolar depression, data from STEP_BD suggest otherwise. In a pragmatic 26 weeks RCT, 179 bipolar depressed patients were randomised to receive adjunctive antidepressants (paroxetine and bupropion) or placebo together with the mood stabilizers they were already on. The main outcome analysed was 8weeks consecutive euthymia. Nearly 24% of the antidepressant group achieved primary outcome while 27% of the placebo group achieved the same.10% in both groups had a ‘switch' to mania (Sachs et al. 2007).

Tricyclic antidepressants carry a greater risk of precipitating a switch to mania than other antidepressants. As depressive episodes tend to be shorter than in unipolar disorder antidepressants may be discontinued after 3-4 months.
Antidepressant-associated chronic irritable dysphoria develops in some bipolar patients who have received antidepressants over long periods; it may be the bipolar equivalent of the antidepressant tachyphylaxis ('poop-out') seen in unipolar depression. A triad of irritability, middle insomnia (restless sleep or night-time awakening), and dysphoria is often seen. Discontinuation of the antidepressant usually results in slow resolution of these symptoms without any worsening of the depressive symptoms (El-Mallakh & Karippot, 2006)
Vigabatrin is an antiepileptic with no proven efficacy in bipolar. It is associated with visual field defects as a side effect. It can also cause psychosis. Similarly, topiramate and phenytoin have no place in the treatment of bipolar disorder.

What is a mood stabiliser?
There is no clear definition for a mood stabiliser. Antiepileptics and lithium are commonly called as mood stabilisers. Bauer et al. proposed that an agent be considered a mood stabilizer if it has efficacy in each of four distinct uses: 1) treatment of acute manic symptoms, 2) treatment of acute depressive symptoms, 3) prevention of manic symptoms, and 4) prevention of depressive symptoms. According to their analysis, only lithium was eligible to be called a mood stabiliser.
Suicide prevention:
For hospital admitted patients, the suicide risk is around 10% over long-term follow-up. Based on naturalistic comparison of patient cohorts, the findings from different centres
Consistently report that lithium could have a suicide prevention effect. The treatment effect is very large for this finding. (Geddes 2003).

Cognitive behavioural therapy for bipolar disorder includes following components:
° Psycho-education. ° Self-monitoring ° Self-regulation: action plans and modification of behaviours ° Increased compliance

Children and adolescents: NICE recognises the possibility of bipolar disorder in those under 18; NICE encourages the use of adult diagnostic criteria except that:

- Mania must be present for diagnosing BPAD
- Euphoria must be present most days, most of the time (for 7 days)
- Irritability is not a core diagnostic criterion. It can be helpful in adolescents if it is episodic, severe, impairs function and is not in keeping or out of character.

3. Schizophrenia & Psychotic disorders

Epidemiology:

Incidence In 1986, WHO published results from 7 countries; ICD 9 schizophrenia was **16 to 42 per 100,000**. Using narrow criteria, it was 7 to 14 per 100,000. It showed considerable between sites variation. McGrath et al. has showed a five-fold difference in the incidence rates of schizophrenia across various sites. Being urban born increases the risk of schizophrenia two-fold compared to rural born individuals.

Living in the city is also noted to increase the incidence of schizophrenia. The incidence of schizophrenia is **3 to 5 times more common in migrants** than native population; this becomes 1.8 when considering prevalence rates. **Winter/spring birth increases the risk of schizophrenia to a small extent** (RR 1.11); but as prevalence of birth itself is common in winter/spring, 10.5% of all schizophrenia incidences can be attributed to the seasonal birth.

Irrespective of broad or narrow definition, the incidence of schizophrenia has definitely increased in certain urban areas over last 40 years. Boydell et al. (BJP, 2003; 182, 45 -49) demonstrated a large increase in Camberwell between 1965 and 1997 using case record analyses. The male: a female difference in the incidence of schizophrenia is estimated to be around 1.4:1, with more males being diagnosed with the disease. The male excess persists even when factors such as age range and diagnostic criteria are taken into account.

Prevalence The median prevalence of schizophrenia is estimated **4.6/1,000 for point prevalence**, 3.3/1,000 for period prevalence, 4.0/1000 for lifetime prevalence (not 1% as quoted in DSM manual), and 7.2/1000 for lifetime morbid risk (Saha et al., 2004). There were no significant differences between males and females, or between urban, rural, and mixed sites, although **migrants and homeless people had higher rates of schizophrenia** and, not surprisingly, **developing countries had lower prevalence rates** (as previously documented). As it is well known that outcome is better in developing nations, point prevalence must be low as a corollary, as it was shown by Saha et al. Of note, **catatonia was 10% in developing vs. 1% in developed nations**; **hebephrenia was 13% in developed vs. 4% in developing nations.**

AESOP study concluded that **all psychoses are more common in the black and minority ethnic group** compared to White population in Bristol, south-east London and Nottingham (crude IRR, 3.6 [95% CI, 3.0-4.2]). Differences in age, sex, and study center accounted for approximately a quarter of this effect (adjusted IRR, 2.9 [95% CI, 2.4-3.5]). The age-sex standardized incidence rate for all psychoses was higher in Southeast London (IRR, 49.4 [95% CI, 43.6-55.3]) than Nottingham (IRR, 23.9 [95% CI, 20.6-27.2]) or Bristol (IRR, 20.4 [95% CI, 15.1-25.7]).

Surveys of psychotic symptoms in general population

ONS 2000 Psychiatric morbidity survey from apparently healthy individuals in private households revealed that 5.5% endorsed at least one psychosis item in a validated questionnaire. 4.2% endorsed hallucination item (Johns, LC BJPsych 2004 185: 298-305) This 4.2% of the sample said that there had been

times when they heard or saw things that other people could not, but only 0.7% reported hearing voices saying quite a few words or sentences when there was no-one around that might account for it.
While 4% of the White British sample endorsed a hallucination question, hallucinations were 2.5-fold higher in the Caribbean sample and half as common in the South Asian sample. (Johns et al., 2002).

	Gender	**Migration**	**Urbanicity**	**Trend**	**Economic status**	**Latitude**
Core Incidence	M>F	Migrant> Native	Urban> Rural/Mixed	Falling over time	Developed = Less developed	High > Lower latitude in males
Prevalence	M=F	Migrant> Native	Urban = Rural/Mixed	Stable	Developed>Less developed	High > Lower latitude
Standardized Mortality (All causes)	M=F	-NA-	-NA-	Rising over time	Developed = Less developed	-NA-

McGrath et al. Epidemiol Rev 2008;30:67–76

<u>High-risk prediction studies in schizophrenia:</u>

- It is thought that the individuals at enhanced genetic risk of schizophrenia (family history) inherit a state of vulnerability characterised by transient and partial psychosis-like symptoms, but not all of these develop florid schizophrenia.
- According to Edinburgh High Risk Study (Johnstone et al., 2005), the 10% risk present in those with high risk family history increases to nearly 50% in those subgroup who have high score on schizotypal cognition and social withdrawal characterised by introversion and anxiety. Measures of episodic memory are also proposed to be significantly discriminating between those with high risk who develop schizophrenia from those who do not; this is suggestive of temporal lobe dysfunction. It is also shown that temporal lobe volume deteriorate with the passage of time in those with psychotic symptoms (this is controversial).
- In an Australian PACE clinic sample, twenty of 49 ultra high-risk subjects (40.8%) developed a psychotic disorder within 12 months. Some highly significant predictors of psychosis were found: long duration of prodromal symptoms, poor functioning at intake, low-grade psychotic symptoms, depression and disorganization.
- In German sample using Bonn Basic Symptoms criteria, a very high conversion rate was seen in help-seeking individuals in a long follow-up period of nearly 10 years. The Bonn Scale has 5 symptom clusters of which the presence of thought, language, perception, and motor disturbances showed a significantly greater predictive discrimination than any other cluster.
- The NAPLS – North American Prodrome Longitudinal study: The ultrahigh risk UHR criteria predict an early transition to psychosis (with a huge relative risk of 405 compared with the incident rate of psychotic disorders in the general population the predictive power can be enhanced by the use of key variables such as genetic risk, functional impairment, and higher levels of psychopathology at baseline (especially odd beliefs and suspiciousness). Substance use (but not a specific class of substance per se)

also increased the predictive value, but the former 3 variables together nearly doubled the predictive power.

Delusional disorders

Delusional disorder	
Incidence	0.7 – 1.3 per 100 000
Prevalence	24 – 30 per 100 000
Proportion of hospital first admission cases	1 -3%
Mean age of onset	39 years
Sex ratio	1.18: 1 = M:F
From Kendler 1982 & Retterstol 1966	

Outcome of schizophrenia: Very long-term follow-ups generally suggest a more favourable outcome that what is assumed by practicing clinicians**.** In the Iowa 500 study, 186 persons with schizophrenia were **followed for an average of 35 years**. Excluding affective or schizoaffective disorder, **46% of those people with schizophrenia had improved or recovered**. The Bonn Hospital Study (Germany) followed 502 persons with schizophrenia for an average of **22.4 years**. The results were that **22% of the research participants had complete remission of symptoms**, 43% had non-characteristic types of remission (defined as involving non-psychotic symptoms only) and 35% experienced characteristic schizophrenia residual syndromes. Therefore, **65% had a more favorable outcome than would have been expected from clinical experience**. In Chestnut Lodge study, 446 patients were followed for an average of 15 years. **One-third (36%) were recovered (strict recovery criteria) or functioning adequately**. In Vermont longitudinal study, 68% of patients (n=269, follow up average 32 years) who underwent a rehabilitation programme had good functioning as determined by GAF scale.

International Study of Schizophrenia – WHO ISoS (1997) reported on the follow-up analysis of two major WHO incidence cohorts. 1st cohort was part of IPSS – International pilot study of schizophrenia while the 2nd cohort was from the Determinants of Outcome of Severe Mental Disorder and the reduction of disability study (DOSMeD) which followed IPSS with more rigorous method and outcome measurement (n=1202, from 9 countries. In total **52% of persons in the developing countries** (e.g., Columbia, India, Nigeria) were assessed to be in the **"best" outcome category** (defined as a single episode only, followed by full or partial recovery) compared with **39% in the developed countries** (e.g., Moscow, London, Washington, Prague, Aarhus and Denmark). In a 5-year follow-up, 73% of those participants from the developing world were in the best outcome group compared with 52% in the developed world. DOSMeD used more rigorous criteria and followed more than 1,300 patients in 10 countries and, similar to the IPSS, discovered that the **highest rates of recovery occurred in the developing world**. DOSMeD excluded mode of onset being a confounding factor. Differential follow-up rates, differential outcome measures, differential sex and age distribution, diagnostic ambiguities did not confound the above results as proved later by Hopper & Wanderling (Scz Bulletin 2000; 26, 835-46)

Risk factors: Risk factors is a broad term which is more usefully divided into risk indicators, which denote increased risk but are not causative, and risk modifiers which are associated with causation. For example, minor physical anomalies are risk indicators. Maternal influenza, perinatal brain insults, family history, etc. are unmodifiable fixed markers. Genetic risk of schizophrenia decreases as one moves from closer to distant relatives. If both parents have schizophrenia, the risk is around 40 to 50%. Social deprivation and illicit drug use are modifiable causal factors.

The table below is adapted from Harrison (2015) Journal of Psychopharmacology 2015, Vol. 29(2) 85–96. This summarizes the most recent genetic findings in schizophrenia from GWAS and CNV studies.

Locus	Gene	Notes
Single Nucleotide Polymorphisms		
12p13.33	CACNA1C (L-type calcium channel)	Important for neuronal function; mutations cause Timothy syndrome and Brugada syndrome
12q24.11	D-amino acid oxidase	Enzyme that degrades d-serine (NMDA co-agonist)
1q42.2	DISC-1 (Disrupted in schizophrenia-1)	Seen in a Scottish family with 1:11 translocation
11q23.2	Dopamine D2 receptor	Target for antipsychotic action
2q33-34	Receptor tyrosine kinase erbB4	Neuregulin 1 receptor
5q33.2	AMPA receptor subunit 1	Affects synaptic plasticity
16p13	NMDA receptor subunit 2A	Influences channel conductance and synaptic localisation
7q21	Metabotropic glutamate receptor 3	Inhibitory autoreceptor
1p21	Micro RNA 137	Regulates transcription
8p12	Neuregulin 1	Growth factor
17p13	Serine racemase	Synthesizes d-serine from l-serine
18q21	Transcription factor 4	Deletion causes Pitt-Hopkins syndrome
2q32	Zinc finger 804A	Affects gene regulation esp. in cortical pyramidal neurons
Copy Number Variations		
2p16.3 deletion	Neurexin 1	Involved in synaptic structure
7q36.3 duplication	Vasoactive intestinal peptide receptor 2	Regulates synaptic transmission in the hippocampus, and development of neural progenitor cells in the dentate gyrus
hemi deletion of 22q11	Involves COMT coding genes	velocardiofacial syndrome

Perinatal complications have stimulated interest as putative risk factors for schizophrenia. Low birth weight was initially implicated as a risk factor; later anoxic brain damage at delivery was implicated, especially the dose of hypoxia was correlated to neuronal death. Kendell in 1996 showed that pregnancy induced hypertension increased the risk of psychosis almost 9 fold in a retrospective study design. Three different types of obstetric events have been described; 1. Growth related events e.g. low birth weight, small for gestational age 2. Perinatal factors e.g. PIH 3. Hypoxic events e.g. premature rupture of membrane. Lewis-Murray scale has been specifically used to measure obstetric complications during childbirth. The odds ratio from various studies has been too low to be significant. It is also not clear if obstetric complications influence age of onset independently; also obstetric complications may be confounded by family history of schizophrenia where difficult perinatal periods may not be uncommon.

Cannabis has been strongly debated in the aetiology of schizophrenia. The interest more or less began with 1960s Swedish conscript study by Andreassen. The pooled estimate for development of psychosis associated with prior cannabis use is an odds ratio (OR) of 2.1 (95% CI: 1.7–2.5). This holds regardless of whether only studies using the narrow clinical outcome are used or whether studies using the broad outcome of psychotic symptoms are considered.

- Confounders such as amphetamine use are not able to explain the association by themselves.
- Reverse causality remains plausible but cannot be proven i.e. patients do not smoke cannabis because they have psychotic mental state or distressed.
- In Dunedin cohort, exposure at age 15 was measured against developing psychosis at age 26 and an association was present even after adjusting for psychotic liability at age 11.
- A Greek cohort study examined the self-medication hypothesis by testing whether subtle psychotic experiences with distress would have stronger associations with cannabis use than psychotic experiences without distress. But stronger associations were found between cannabis and psychotic experiences in the absence of distress, making self-medication theory unlikely. Nevertheless, there are studies that suggest a bidirectional association.
- Cannabis shows dose-response relationships to psychosis outcome; exposure often precedes the outcome, and there is a plausible biological mechanism linking the cannabis and schizophrenia either via role of endocannabinoids in neurodevelopment or dopamine sensitization.

Psychosocial interventions in schizophrenia:

Psychosocial interventions in schizophrenia are aimed at improving functional outcomes and quality of life and not merely improving PANS or SAPS scores.

Psychosocial interventions are classified as obligatory ones and voluntary ones by Bauml et al. The process of empowerment of patients and their relatives to understand and accept the illness and cope with it in a successful manner is described as a basic-level competency. This is considered to be an ''obligatory-exercise'' program upon which additional ''voluntary-exercise'' programs such as individual behavioral therapy, self-assertiveness training, problem-solving training, communication training, and further family therapy interventions can be built.

Thus, **psychoeducation** becomes the most basic and important part of psychosocial interventions in schizophrenia. The term ''psychoeducation'' was first employed by Anderson and was used to describe a behavioral therapeutic concept consisting of 4 elements; briefing the patients about their illness, problem solving training, communication training, and self-assertiveness training, whereby relatives were also included. Cochrane review of psychoeducation demonstrates a reduction in relapse and improved compliance. Psychoeducation does not increase suicidal thoughts in patients. It empowers patients and helps carers to act as cotherapists.

Family Therapy:

Brown & Rutter demonstrated that schizophrenia patients in families with high expressed emotion were more likely to experience a relapse during the following year despite adequate pharmacotherapy. Family therapy reduces relapse rates from nearly 64% to 24% in such situations. The effectiveness of family therapy is more if the baseline relapse risk is increased in the first place. A Cochrane review (Pharaoh 2000) shows than NNT for family therapy in relapse prevention for schizophrenia is around 6.

Social skills training: Following the framework described by Bellack and Mueser, there are three forms of social skills training:

1. The **basic model**: here complex social repertoires are broken down into simpler steps, subjected to corrective learning, practiced through role playing, and applied in natural settings.
2. The **social problem-solving** model: This focuses on improving impairments in information processing that are assumed to be the cause of social skills deficits. The model targets domains needing changes including medication and symptom management, recreation, basic conversation, and self-care.
3. The **cognitive remediation** model: Here the corrective learning process begins by targeting more fundamental cognitive impairments, like attention or planning. The assumption is that if the underlying cognitive impairment can be improved, this learning will be transferred to support more complex cognitive processes, and the traditional social skills models can be better learned and generalized in the community.

CBT in psychosis:

Though CBT is gaining popularity in treating psychotic symptom, clinicians are still unsure about the symptoms targeted in CBT. According to Birchwood, the target is emotional dysfunction that accompanies psychotic experience and not the psychotic symptoms per se. Turkington described the following elements in CBT for psychosis:

1. Therapeutic alliance – not colluding with delusions but validation.
2. Improving medication adherence.
3. Providing alternate explanations to unusual experiences e.g. normalisation.
4. Decreasing the impact of positive symptoms e.g. addressing the omnipotence of voice.
5. Graded reality testing using peripheral questioning and inference chaining.

CBT strongly complements the recovery model for schizophrenia. It is unclear when and how CBT should be delivered, what are the most effective and essential components of such CBT, the reliability of CBT among various therapists, what kind of patient is the most suited, etc.

Some trials have shown reasonable benefits for the first episode of psychosis using CBT. Overall considering CBT in psychosis, thirty-four trials were included in a meta-analysis which concluded positive beneficial effects for the target symptom (33 studies; effect size = 0.4) as well as significant effects for positive symptoms (32 studies), negative symptoms (23 studies), functioning (15 studies), mood (13 studies), and social anxiety (2 studies) with effects ranging from 0.35 to 0.44. However, there was no effect on hopelessness. Trials in which raters were aware of group allocation had an inflated effect size of approximately 50%–100%. Secondary outcomes (e.g., negative symptoms) were also affected, but the effect sizes were not significant.

Vocational rehabilitation:

- Competitive employment has been estimated at less than 20% for severely mentally ill persons and is probably lower in patients with schizophrenia.
- Traditional approaches include vocational training, voluntary working, clubhouses run by mental health service providers, and sheltered employment.
- Supported employment programs, aims to improve opportunities for competitive employment. This follows individual placement model where place- and- train sequence is used instead of the train- and -place concept. Several common components across various models of supported employment are 1. including a goal of permanent competitive employment 2. minimal screening for employability, 3. avoidance of preoccupational training, 4. individualized placement (i.e., not enclaves or mobile work crews), 5. time-unlimited support, 6. consideration of client preferences

Three randomized controlled trials for supported employment programs that had competitive employment as the primary targeted outcome were analysed, and significant advantages for supported employment programs over control interventions have been demonstrated. The unweighted mean of patients in supported employment programs for obtaining competitive employment was 65% whereas the corresponding rate for patients in the control conditions was 26%.

Prognosis and recovery in schizophrenia:

Relapse & recovery after a psychotic episode

27% of patients with a first episode of psychosis have a relapse within one year irrespective of antipsychotic treatment.
61% of patients with a first episode of psychosis have a relapse within one year if receiving placebo and not antipsychotics.
48% of patients with five or more previous episodes relapse within one year irrespective of antipsychotic treatment
87% of patients with five or more previous episodes have a relapse within one year if receiving placebo and not antipsychotics.
40% of all patients relapse within one year in spite of taking antipsychotic drugs;
62% of patients living in a stressful environment relapse within one year in spite of taking antipsychotic drugs.
19% relapse within one year if antipsychotics and family education are combined;
20% relapse within one year if antipsychotics and social skills training are combined;
Recovery at 15-25 years as defined by GAF>60 is seen in 37.8% of patients with schizophrenia and 54.8% of patients with other psychoses

Adapted from **Byrne, P. Managing the acute psychotic episode. BMJ, 2007; 334(7595): 686 - 692.**

Suicide:

A meta-analysis of 61 studies by Palmer et al. has estimated lifetime suicide prevalence in those observed from first admission or illness onset with schizophrenia as 5.6% (95% confidence interval, 3.7%-8.5%). Most of these suicides occur within few years of the illness onset. Non-fatal acts of self-harm are also increased, with a study of people with chronic schizophrenia finding that 38% had at least one episode of self-harm in a 2- to 12-year follow-up period.

Mortality rates:

The **median standardised mortality rate for schizophrenia** patients was **2.58**, with no obvious sex difference. **Suicide was associated with the highest SMR (12.86)**; however, most of the major causes-of-death categories were found to be elevated in people with schizophrenia. The SMRs for all-cause mortality have increased during recent decades (*P* = .03). Considering that (1) case fatality rates (i.e. 'deadliness' of schizophrenia) for schizophrenia did not significantly differ among the decades and (2) standardized mortality rates are generally decreasing in most nations, increasing SMR in schizophrenia suggests that people with schizophrenia have not fully benefited from the improvements in health outcomes available to the general population. As mortality rates in the general population decreased over time at a faster rate than those for people with schizophrenia, the SMRs for people with schizophrenia have increased over time. This suggests that the differential mortality gap has widened over time. Of note is the fact that the SMRs in schizophrenia did not differ by economic status.

Prognostic predictors:

The prognostic significance of schizophrenia subtypes employed in current classificatory systems is controversial. In fact, well designed longitudinal studies exploring the prognostic differences among the subtypes are scarce. Despite this DSM-IV text revision asserts that these subtypes have sufficient prognostic value to be retained. The paranoid and catatonic subtypes of schizophrenia have the best outcome; disorganised (hebephrenic) subtype has the worst prognosis. The undifferentiated type has an intermediate prognosis. Subtyping and rediagnosis of 187 schizophrenic patients from the Chestnut Lodge follow-up study (19 years long) throws some light on this issue. Paranoid schizophrenia had an older age at onset, often developed rapidly in individuals with good premorbid functioning, tended to be intermittent during the first 5 years of illness, and was most associated with good outcome or recovery. Hebephrenia had an earlier age at onset, often developed insidiously, and was associated with a greater family history of psychopathology, poor premorbid functioning, and, frequently, a continuous illness with a poor long-term prognosis. While also early and insidious in onset, unlike hebephrenia, undifferentiated schizophrenia was poorly distinguished from the patients' premorbid state, associated with an early history of behavioral difficulties, and often resulted in a continuous but stable disability (intermediate between paranoid and hebephrenic).

Prognostic factors in Schizophrenia (with respect to relapses and rehospitalisations)	
Good Prognosis	**Poor Prognosis**
Late onset	Early onset* (not strong)
Obvious precipitating factors	No precipitating factors**
Acute onset	Insidious onset**
Good premorbid adjustment	Poor premorbid adjustment
Affective symptoms (esp. depression)	Social withdrawal
Being married	Single, divorced, or widowed
Family history of affective disorders	Family history of schizophrenia* (weak, controversial)
Good social support	Poor social network, High EE families
Positive symptoms only	Negative symptoms
Good initial response to treatment (best predictor)	Poor compliance
	Neurological signs and symptoms
	History of perinatal trauma
	No remissions in 3 years
	Many relapses
	History of violence
* Suvisaari, JM. Br J Psychiatry. 1998 Dec;173:494-500. ** Rosen, K & Garety, P. *Schizophr Bull* . 2005; 31:735–750.	

Adapted from Kaplan and Sadock's Synopsis of Psychiatry, 9th ed. New York, NY: Lippincott Williams & Wilkins; 2003.

Robust predictors of short-term outcome (in an acute episode):
Stressful life events, high expressed emotions and noncompliance are the best predictors of poor response to a psychotic episode.

Robust predictors of medium (2 to 5 years) outcome:
A good outcome is seen in females, married patients, those with social contacts outside the home, and in acute onset. Age of onset is not a consistent predictor in prognostic studies. Family history of psychosis does not consistently predict the worse outcome (some studies showed better outcome in those with higher familial loading). Clear-cut negative symptoms that appear early in the course have some predictive power for poor prognosis. The course of illness in first 2 years is the best predictor of the subsequent course. First rank symptoms have no prognostic significance.

Evidence-based treatment of schizophrenia

RCT evidence to support practice in schizophrenia is analysed under following areas: (Davis & Leucht, 2008)

1. Do drugs work?

There is excellent evidence that both first-generation antipsychotics (FGAs) and second-generation antipsychotics (SGAs) substantially benefit patients better than placebo alone based on over 30 double-blind studies. There are also a number of comparisons of second-generation drugs vs. first-generation drugs.

2. Choice of drugs:

It is only clozapine that shows moderate to large effect size differences from FGAs on clinical outcome. Olanzapine and risperidone show low to moderate effect size differences from FGAs, but not consistently (Tamminga & Davis 2007).No evidence to suggest which one antipsychotic must be used in which clinical situation while treating schizophrenia. Clozapine has evidence in resistant schizophrenia – not first episode. CATIE and CUtLASS showed that second-generation drugs were no better than first generation ones in terms of efficacy and cost. Olanzapine fared better than other atypicals in CATIE in terms of dropouts. (see below at research update section for further info on CATIE)

3. Dose:

There is no evidence as to the correct mean dose for FGA drugs, but there is some evidence for SGAs. There is no information about when to escalate or reduce the dose. There is a limited evidence study that shows first-episode patients require less medication. But there are no RCTs of first-episode patients randomized to several doses. Dose escalation with chronicity or severity has not been studied in RCTs. There is no evidence that a lower dose is required for maintenance than for the acute episode.

In a meta-analysis of dose comparison studies (Davis & Chen 2002), it was determined that for typical antipsychotic haloperidol the near maximal effective dose ranges from 3.3–10 mg/day. There was no evidence that higher doses of first generation antipsychotics were more effective than medium doses (3.3–10 mg/day haloperidol or equivalent). For atypical antipsychotics, the near maximal efficacy dose for aripiprazole was 10 mg/day, clozapine >400 mg/day, olanzapine possibly >16 mg/day, and risperidone 4 mg/day. (For risperidone, a dose of 2 mg daily consistently produced a lower level of efficacy. Doses of 6 mg or greater produced no additional benefit and doses tend to be less efficacious at 10 mg daily and above. (Ezewuziie & Taylor, 2006)). The optimal dose of risperidone in relapsed schizophrenia is 4mg daily. Higher doses are unlikely to improve efficacy and may reduce it. Adverse movement disorders become more common.For both atypical and typical antipsychotics, there was no evidence that higher doses were less effective, arguing against a "therapeutic window". The Royal College guideline on high dose prescription is discussed below.

Chlorpromazine	100mg/day
Risperidone	2 mg/day
Olanzapine	5 mg/day
Quetiapine	75 mg/day
Ziprasidone	60 mg/day
Aripiprazole	7.5 mg/day.
Haloperidol	2mg/day
Woods SW. Chlorpromazine equivalent doses for the newer atypical antipsychotics. J Clin Psychiatry. 2003 Jun;64(6):663-7.	

4. Emergency treatment:
There is no evidence base to suggest whether benzodiazepines alone or combination with antipsychotics will be needed for rapid tranquillisation of the acute agitated psychotic patient.

5. When to change drug or augment:
RCTs have not informed us as to when, with what drug, and in which patient augmentation is useful. There is some evidence to indicate that patients who seem to have depressive disorders superimposed on schizophrenia may respond to augmentation with an antidepressant or, if recurrent, need prophylactic antidepressants.

Essock et al. (2007) used the CATIE data to find out whether switching or staying with same regimen will be useful in those who require a review of their treatment. In general, the "stayers" did better than the "switchers" for ***each medication***, but especially for olanzapine. Thus, if patients are taking olanzapine or risperidone and switch to another medication, they will probably have a poorer outcome than if they did not switch.

6. Specific situations:

A. Aggression:

A 12 weeks double-blind RCT compared clozapine, olanzapine, and haloperidol in high-risk patients with a history of aggressive behaviour. Clozapine showed greater efficacy than olanzapine and olanzapine greater efficacy than haloperidol in reducing aggressive behaviour. This anti aggressive effect appeared to be separate from the antipsychotic and sedative action of these medications.

B. Negative symptoms:

- Negative symptoms include anhedonia, avolition, apathy, amotivation and restricted affect. The term ‘negative symptoms’ is a descriptive term.
- Primary and secondary negative symptoms refer to distinct subgroups of negative symptoms differing in their cause, longitudinal stability, or duration and differentiated through longitudinal observation.
- **Primary** and enduring **negative symptoms** or deficit symptoms refer to the symptoms that are intrinsic to schizophrenia.
- **Secondary negative symptoms** refer to negative symptoms occurring in association with, or presumably caused by, positive symptoms, affective symptoms, medication side effects, environmental deprivation, or other treatment- and illness- related factors.
- In clinical samples, patients with the deficit form of schizophrenia or primary negative symptoms represent about 20%–30% of patients, whereas in population-based samples approximating incidence samples, patients with the deficit form of schizophrenia comprise 14%–17% of patients with schizophrenia.
- The **deficit schizophrenia** is defined by the following criteria:

1. At least 2 of the following 6 features must be present and of a clinically significant severity: (1) restricted affect (2) diminished emotional range (3) poverty of speech (4) curbing of interest (5) diminished sense of purpose and (6) diminished social drive.
2. Two or more of these features must have been present for the preceding 12 months and must have always been present during periods of clinical stability (including chronic psychotic states). These symptoms may or may not be detectable during transient episodes of acute psychotic disorganization or decompensation.
3. Two or more of these enduring features are also idiopathic, i.e., not secondary to factors other than the disease process. Such factors include: (1) anxiety, (2) drug effects (especially, extrapyramidal side effects), (3) suspiciousness, (4) formal thought disorder, (5) hallucinations or delusions, (6) mental retardation, and (7) depression.
4. The patient meets DSM-5 Schizophrenia criteria. Note that in DSM-5, as opposed to DSM-IV, there is no special attribution to bizarre delusions or first-rank auditory hallucinations. In DSM-5 two Criterion A symptoms are required for any diagnosis of schizophrenia (among them at least one of: delusions, hallucinations and disorganized speech).

- Various features are suggested to be associated with deficit syndrome:

Deficit syndrome - associations
Frontal atrophy
Familial pattern
Summer birth
Increased eye tracking dysfunctions
More tardive dyskinesia
Poorer functional outcome
Lower suicide and depression rates

- There are 3 studies that compared the efficacy and tolerability of antipsychotics in deficit and nondeficit subjects. Two of them looked into olanzapine while the third looked into clozapine. Unfortunately, no significant improvement in deficit symptoms was noted in all 3 trials.
- SGAs are marketed as being especially efficacious for negative symptoms; good evidence to support this hypothesis is scarce. Most RCTs are conducted on patients with predominantly positive symptoms.
- **Amisulpride** is still the best-examined SGA as regards negative symptoms because several studies in patients with predominantly negative symptoms have been carried out with this compound. But the effect of amisulpride in reducing negative symptoms at a low dose (50 to 100mg/day) may be due to a concurrent reduction in positive symptoms or depression. When compared head to head with haloperidol, the effect of amisulpride on negative symptoms weakened.
- A well-conducted study appeared to show superiority for **Olanzapine** (only 5mg/day) over Haloperidol, but the magnitude of the effect was modest.
- Efficacy of add-on therapy with d-cycloserine, a partial agonist acting at the glycine modulatory site of the glutamatergic N-methyl-d-aspartate receptor, was examined in 3 double-blind trials in patients with deficit syndrome. 2 of the 3 studies suggest add-on d-cycloserine 50 mg/day may have some effect on negative symptoms in patients with deficit syndrome despite small sample sizes and different study designs and duration.

- Glycine administration resulted in a significant 30% 6 16% reduction (P < 0.001) in negative symptoms in a study wherein positive symptoms were also concurrently reduced.
- Selegiline was studied in a small number of patients in a 12-week placebo-controlled trial with some favourable results.

C. Suicide prevention:

In terms of suicide prevention, the Intersept study (INTERnational SuicidE Prevention Trial) compared clozapine and olanzapine. This was a 2-year multicenter RCT study comparing the risk for suicidal behaviour in 980 patients with schizophrenia or schizoaffective disorder treated with clozapine vs. olanzapine. Suicidal behaviour was significantly less in patients treated with clozapine vs. olanzapine (hazard ratio, 0.76; 95% confidence interval, 0.58-0.97; P =.03).

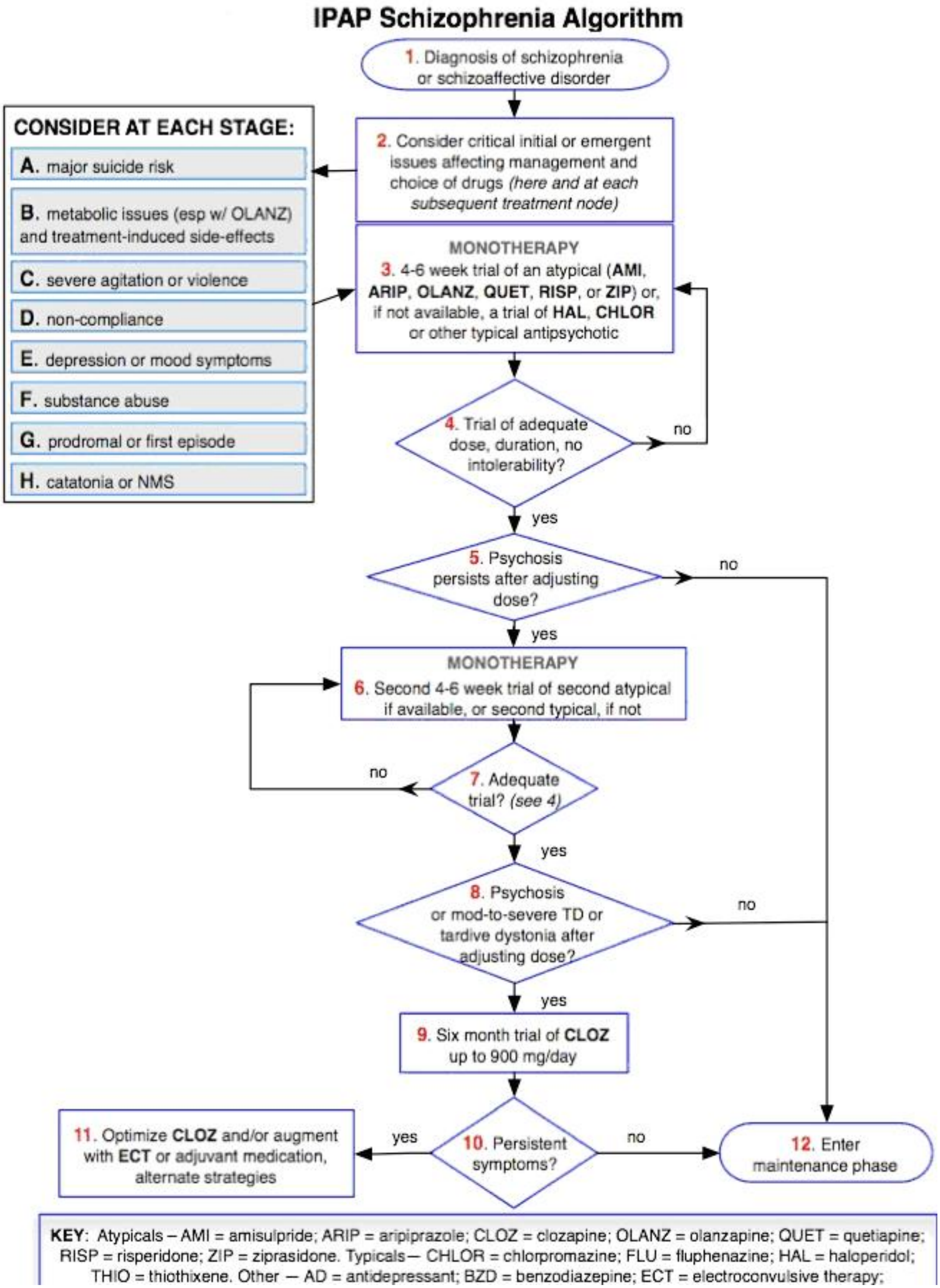
D. Cognitive symptoms:

Cognitive symptoms occur independent of negative symptoms in schizophrenia. A number of studies have claimed cognitive benefits from treatment with various second-generation antipsychotic drugs (SGAs). This claim is based on randomised comparative trials that demonstrated that SGAs produce significant improvement in neurocognition, measured by means of cognitive tasks administered at baseline and follow-up assessments. Goldberg et al. 2007 designed an RCT of risperidone vs. olanzapine in patients with first-episode schizophrenia, including a control group of individuals for whom data from multiple assessments were also collected. The main finding was that the effect size for cognitive change in patients exposed to SGAs was very similar to the effect size for cognitive change calculated in the healthy control group. Hence, cognitive improvements observed in RCTs may have been due to the non-specific effects of multiple testing (practice effects).

7. Treating the first episode:

Summary of NICE guidelines (from Patel & David 2005):
The choice of antipsychotic drug should be made jointly by the patient and responsible clinician, on the basis of an informed discussion of benefits and side-effects
Oral atypical antipsychotics are recommended as first-line treatment for patients with newly diagnosed schizophrenia
If a patient on oral typical antipsychotics has adequate symptom control but is experiencing unacceptable side-effects, an oral atypical should be considered
If a patient on an oral typical has good symptom control and no unacceptable side-effects, a routine switch to an atypical preparation is not recommended
Clozapine should be used at the earliest opportunity for patients with evidence of treatment-resistant schizophrenia
A risk assessment should be performed regarding treatment adherence, and depot preparations should be prescribed when appropriate
Where more than one atypical drug is considered appropriate, the drug with the lowest purchase cost (allowing for daily required dose) should be prescribed

Where full discussion between the patient and responsible clinician is not possible, oral atypicals should be the treatment of choice because of the lower potential risk of extrapyramidal symptoms
Antipsychotic therapy should be initiated as part of a comprehensive package of care that addresses the patient’s clinical, emotional and social needs
Atypical and typical antipsychotics should not be prescribed concurrently, except for short periods to cover changeover of medication



From **www.ipap.org/schiz**

Note that other factors such as risk posed by the patient, support available and presence of comorbidities could dictate differences in treatment and admission to acute wards.

The treatment of **later episodes** will depend on: compliance with medication, therapeutic response, side-effects, the cause of relapse and treatment resistance.

8. Maintenance phase:

The antipsychotic medication must be continued at the minimum necessary dose. Often this is the same dose as the dose to which the acute phase responded. The goals for maintenance treatment are to prevent psychotic relapse and to assist patients in improving their level of functioning. 20% of patients receiving treatment will relapse within a year despite treatment; 60% will relapse if not taking medications. Even those with only one episode have a four in five chance of relapsing at least once over the following 5 years. Stopping medication increases this risk fivefold. Most guidelines suggest that 1 or 2 years maintenance is needed after an episode; this is arbitrary and may not be adequate. It is generally recommended that multiepisode patients receive maintenance treatment for at least 5 years after last episode, and if needed, on an indefinite basis.

Switching during maintenance:

PROBLEM	SWITCH TO
EPSEs (acute EPSEs such as dystonia, parkinsonism are measured using Simpson-Angus scale; tardive dyskinesia can be measured using AIMS scale)	Atypicals; avoid a higher dose of any atypical, especially risperidone.
Metabolic syndrome (weight gain, diabetes, high lipids, high BP)	Avoid clozapine and olanzapine; Amisulpride and aripiprazole may be better
High prolactin	Rule out other causes; Aripiprazole, olanzapine and quetiapine relatively safe. Avoid amisulpride
QT prolongation	No clear recommendations; olanzapine and aripiprazole may be useful
Sedation	Haloperidol, aripiprazole and amisulpride preferred
Tardive dyskinesia	Lesser with clozapine and other atypicals.
Sexual dysfunction	No specific relief but aripiprazole and quetiapine may have a better profile.

Adapted from Maudsley prescribing Guidelines

Depot medication:

A meta-analysis (Adams et al.) found greater global improvement among patients given depot antipsychotics than among those given oral antipsychotics and found similar rates of side effects in the two groups.

In a survey of outpatients by Pereira & Pinto (1997), 87% of those receiving depot antipsychotics, given a free choice, elected to continue with their present dose form (depots). More than 60% of patients converted to depot medication preferred the injection to their previous tablet treatment and said that they felt better during the injection regimen. In a qualitative survey, patients on depot felt that 'more normal lives' are possible and that depots were a safety net protecting them from relapses and rehospitalisations

Depot	Characteristic features
Zuclopenthixol	? More effective in aggressive patients. May be better than other depots in relapse prevention. EPSE burden high.
Flupentixol	May have an antidepressant effect; can be given at a higher chlorpromazine equivalent dose but still within BNF limits
Haloperidol	Useful in the prevention of manic relapses too. May need 3-6 months to reach steady state
Pipotiazine	Fewer incidences of EPSEs.
Fluphenazine	May induce depressed mood
Risperidone depot	Need aqueous suspension immediately before injection; needs to be stored in fridge and test dose is not required.
Flupenthixol decanoate 40mg/2wks = Fluphenazine decanoate 25mg/2wks = Haloperidol decanoate 100mg/4wks = Pipothiazine palmitate 50mg/4wks = Zuclopenthixol decanoate 200mg/2wks Risperidone 25mg/2wks depot is equivalent to approx. 4mg/d tablets	

Adapted from **Maudsley prescribing Guidelines**

Tackling myths about depot drugs (from Patel & David, 2005; Carpenter & Buchanan 2015)

- The risk of neuroleptic malignant syndrome is not higher for depot than oral drugs
- There is no evidence to suggest that neuroleptic malignant syndrome is a contraindication for subsequent depot use
- For the same drug, the risk of tardive dyskinesia is not higher for depot than oral formulations
- Patients already on depot like this formulation and many prefer depot to oral drugs
- Although depots are sometimes associated with coercion and reduced patient autonomy, coercion is not very common
- Clinicians perceive a higher level of stigma being associated with depots, but this may be based on the worst characteristics of typical drugs (e.g. unacceptable side-effects) rather than on intramuscular long-acting injections *per se*
- Most nursing staff are aware of the benefits of depots, but their training experiences and pressure of time may adversely affect systematic monitoring of potential side-effects
- Depot clinics are relatively cheap to run, and the financial benefits of avoiding rehospitalisation are clear-cut, but acute services currently have preference in terms of service planning and budgeting

Royal College of Psychiatrists guidelines for high-dose prescribing:

- High-dose antipsychotic prescribing (>1 g chlorpromazine or equivalent) should be avoided wherever possible. There is no evidence that such high dose prescribing provides symptom relief in those who have not responded to monotherapies or combinations of low doses.
- Consider alternative approaches including adjuvant therapy and newer or atypical antipsychotics such as clozapine when considering high-dose prescription.
- Risk factors including metabolic effects may be more pronounced in those greater than 70.
- Consider the potential for drug interactions at high doses and combinations.

- Carry out ECG to exclude abnormalities such as prolonged QT interval; repeat ECG periodically and reduce dose if prolonged QT interval or another adverse abnormality develops.
- Increase dose slowly and not more often than weekly.
- Regular physical examination including vital signs and hydration status must be monitored.
- Consider high-dose therapy for a limited period only; there is no use continuing after 3 months if no response has occurred.

9. Treatment-resistant schizophrenia:

- Pivotal evidence for the use of clozapine in treatment-resistant schizophrenia comes from Kane et al., (1988) and a meta-analysis of randomised controlled trials of clozapine (Wahlbeck et al., 1999).
- Kane et al. reported a multicenter clinical where schizophrenia patients who had failed to respond to at least three different neuroleptics underwent a prospective, single-blind trial of rather a high dose of haloperidol for six weeks. Patients whose condition remained unimproved in spite of haloperidol were then randomly assigned, in a double-blind manner, to clozapine (up to 900 mg/d) or chlorpromazine (up to 1800 mg/d) for six weeks. Out of 268 patients, 30% of the clozapine-treated patients were categorized as responders compared with 4% of chlorpromazine-treated patients. This improvement included "negative" as well as positive symptoms.
- Meltzer concluded that 30% of clozapine receivers would respond by 6 weeks, a further 20% by 3 months and an additional 10–20% by 6 months. The improvement rates may continue up to 1 year. Therefore, it seems reasonable to try clozapine monotherapy at least for 6 months.
- There is no meaningful relationship between clozapine plasma levels and clinical response. However, there is a consensus in the literature that a plasma level of about 350–450 ng/ml has to be attained before the patient is considered to be non-respondent to clozapine (Kerwin & Bolonna 2005).
- Psychopathology cannot predict treatment response to clozapine. Presence or absence of first rank symptoms does not identify clozapine responders.
- Conley et al. (1998) reproduced the methodology of the pivotal Kane et al.(1988) study of clozapine but concluded that olanzapine was no better than chlorpromazine in treatment resistance.
- NICE recommends using clozapine at the earliest in treatment resistance: defined arbitrarily as lack of satisfactory clinical response to sequential use of at least two antipsychotics for a period of 6 to 8 weeks: at least one of the agents must be from atypical (non-clozapine) group. Please note that other groups may define Treatment Resistance differently.

Clozapine resistance could be a potential problem.

- o Augmentation with risperidone, fluoxetine (possibly pharmacokinetic via CYP inhibition) and anticonvulsants have been studied.
- o Augmentation strategies incorporating sulpiride and amisulpride are also useful as a high potency D2 blockade is possible with these drugs. This is not a pharmacokinetic effect as clozapine levels remain unaffected.
- o The therapeutic effects of lamotrigine augmentation were assessed in 34 clozapine-resistant patients over 14 weeks in an RCT cross-over study(Tiihonen et al., 2003). Lamotrigine treatment significantly

improved positive symptoms and general psychopathological symptoms, but had no effect on negative symptoms.

- o A review of risperidone augmentation provides some evidence when mean clozapine dose is around 475mg/day and mean risperidone dose is around 4.5mg/day. A lower risperidone dosage and a longer duration of the trial seemed to be associated with a better outcome (Kontaxakis, 2005)
- o There is some evidence that ethyl-eicosapentanoate (fish omega oil) may serve as an adjunct to clozapine.

CATIE summary

- CATIE stands for Clinical Antipsychotic Trials of Intervention Effectiveness.
- The study design was double-blind pragmatic RCT.
- 1493 patients with chronic schizophrenia (mean duration of illness = 14 years), 57 sites, 2001 to 2004
- Olanzapine, quetiapine, Risperidone, ziprasidone (added later in the trial), perphenazine
- Primary outcome is a 'real-world' measure – discontinuation for any reason, either patient initiated or physician initiated
- 76% power to detect 12% difference in primary outcome
- Irrespective of the prescribed drug – 74% discontinued treatment in 18 months (surprisingly high despite naturalistic design). Median time to discontinue was 4.6 months.
- Olanzapine had lowest discontinuation rate (still 64%) – but highest side effect burden. 64% discontinued olanzapine; 75%, perphenazine; 82%, quetiapine; 74%, risperidone; and 79%, ziprasidone.
- Olanzapine caused most weight gain while quetiapine caused most anticholinergic symptoms, perphenazine had highest EPSE related discontinuation.
- Those who did not respond after 18 months (those who discontinued for the ineffectiveness of therapy) were re-randomised in phase 2 trial (n=99), and Clozapine was compared against other atypical agents (efficacy pathway). Clozapine had lowest discontinuation rate – median at 10 months. This time-to-discontinuation was nearly 3 times longer than time-to-discontinuation with the other SGAs.

As a part of the phase 2 CATIE study (tolerance pathway) those who terminated phase 1 for ''intolerable side effects'' (444 volunteers) were tested with olanzapine, risperidone, quetiapine, or ziprasidone. Of these treatments, olanzapine and risperidone had equivalent effectiveness, and both were better than quetiapine or ziprasidone by significant but modest margins.

CATIE Controversies

- o Quiet complicated study design and many outcomes were analysed from the dataset.
- o Decisions to add ziprasidone to the protocol was made after recruitment began
- o Perphenazine was used only in one randomized phase (phase 1) of the study generating controversy.
- o The decision to use double-blinded treatments decreased the resemblance of the study procedures to those of routine clinical care
- o The mean doses used remain controversial though it is claimed that the study was designed to be pragmatic and not purely experimental.

Drug	Median dose used
Olanzapine	20.1 mg
Quetiapine	543.4 mg
Risperidone	3.9 mg
Ziprasidone	112.8 mg
Clozapine (phase2efficacy)	332.1 mg
Quetiapine (phase2efficacy)	642.9 mg
Risperidone (phase2efficacy)	4.8 mg
Olanzapine (phase2efficacy)	23.4 mg

CUtLASS summary:

- CUtLASS stands for Cost Utility of the Latest Antipsychotic Drugs in Schizophrenia Study
- It is an unblinded randomised controlled trial comparing first-generation *v.* second-generation antipsychotics
- The primary outcome was the quality of life at 1 year and symptom measures were the main secondary outcome.
- 1, 227 people with schizophrenia who were being assessed by their clinical team for medication review because of poor response or adverse effects were randomised.
- The second-generation drugs were amisulpride, olanzapine, quetiapine or risperidone.
- The rate of follow-up interview was 81% at 1 year.
- The results showed no advantage of second-generation drugs in terms of quality of life or symptom burden over 1 year with those on first-generation antipsychotic doing relatively better.
- Participants reported no clear preference for either class of drug.
- The second phase - CUtLASS 2 trial was of similar design and compared clozapine with other second-generation drugs in 136 patients who had not responded well to two or more previous drugs. Results showed that there was a significant advantage for clozapine in symptom improvements over 1 year; moreover, patients significantly preferred it.

Treatment of other schizophrenia-like psychotic disorders:

Delusional disorders:

- In a triple blinded RCT comparing efficacy of fluoxetine in body dysmorphic disorder (40% patients had delusional BDD), fluoxetine was increased by 20 mg/day every 10 days to a maximum of 80 mg/day. A significant response started occurring at 8 weeks until 12 weeks when the observation was made. Patients who were delusional at baseline were as likely as non-delusional patients to respond to fluoxetine.

Outcome	Fluoxetine	Placebo	NNT (CI)
≥30% decrease on BDD-YBOCS score	53%	18%;	3 (2 to 9)

- A systematic review failed to identify any RCTs in treating primary delusional parasitosis. Various heterogeneous case reports have used both atypicals and typicals, and no recommendation could be made on the basis of available evidence.

Post schizophrenic depression:

- Levinson conducted a systematic review and concluded that antidepressants are beneficial for treatment of depression in people with schizophrenia who are stable with respect to psychotic symptoms, whereas antipsychotics were more effective in people with acute psychosis.
- The occurrence of depression in chronic phase may respond to atypical antipsychotics themselves; any associated depressive symptoms during the acute phase of psychosis respond well to neuroleptics/atypicals.
- Note that both clozapine (vs. olanzapine) and olanzapine (vs. haloperidol) have antisuicidal effects in schizophrenia.
- The role of antidepressants is not clearly proven though many clinicians use antidepressants as adjuvants in schizophrenia with depression.

Schizoaffective disorder:

- Mood stabilizers are a mainstay of treatment for patients with schizoaffective disorder.
- Carbamazepine is superior for schizoaffective disorder, depressive type; lithium and carbamazepine have no differences in efficacy for the manic schizoaffective (bipolar like) disorder.
- Antipsychotics are often used as combination medications with mood stabilisers or primarily for mood stabilising effect itself.
- Treatment with antidepressants is similar to the treatment of bipolar depression.

Psychotic depression:

- A metaanalysis of 10 eligible RCTs by Wijkstra et al. showed that the combination of antidepressant and antipsychotic is no better than antidepressant monotherapy. However, the combination was better than placebo; antidepressants or antipsychotic monotherapies compared equivalently with neither being superior.

Intervention	Comparison	Result
Antidepressant alone	Antipsychotic alone	No difference
Antidepressant + Antipsychotic	Antidepressant alone	No difference
Antidepressant + Antipsychotic	Antipsychotic alone	Combination superior
Antidepressant + Antipsychotic	Placebo	Combination superior

- In spite of overall high withdrawals, there were no significant differences in withdrawal rates between any of the treatments. American Psychiatric Association and NICE guidelines recommend a combination strategy as first-line treatment in people with psychotic depression.

DISCLAIMER: This material is developed from various revision notes assembled while preparing for MRCPsych exams. The content is periodically updated with excerpts from various published sources including peer-reviewed journals, websites, patient information leaflets and books. These sources are cited and acknowledged wherever possible; due to the structure of this material, acknowledgements have not been possible for every passage/fact that is common knowledge in psychiatry. We do not check the accuracy of drug-related information using external sources; no part of these notes should be used as prescribing information

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